

Modelos SSE

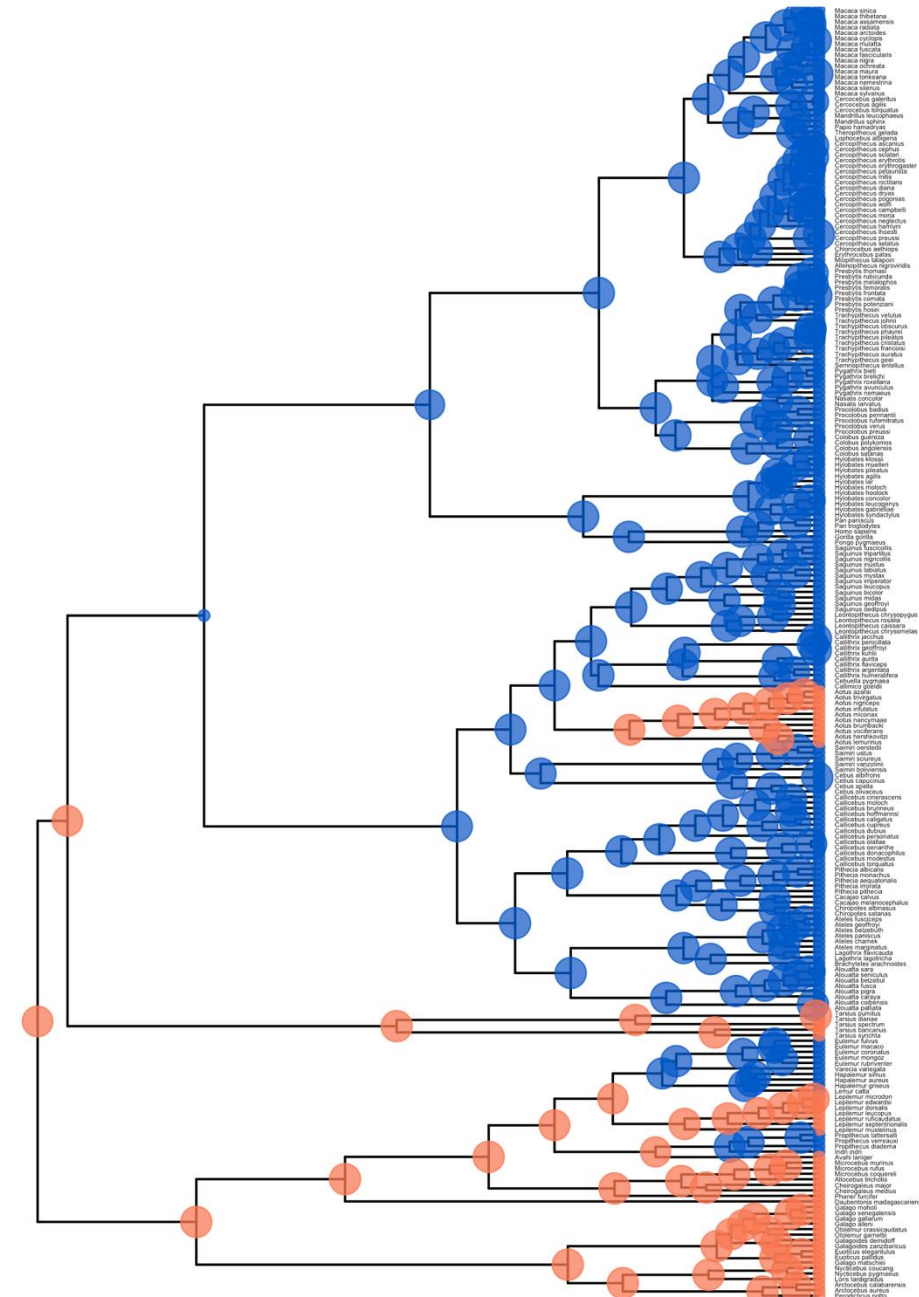
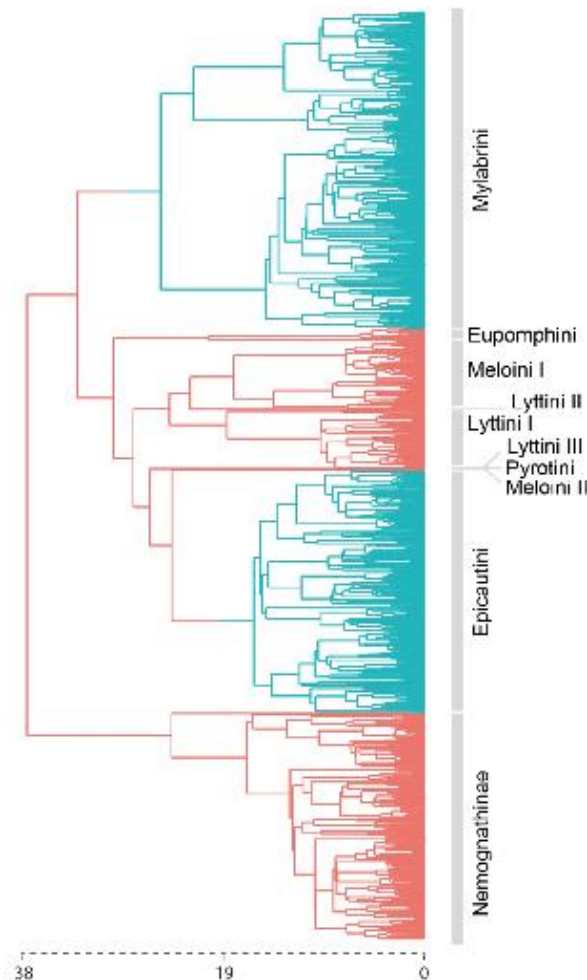
(State-dependent Speciation and Extinction)

CONTENIDO

- 1) La motivación
- 2) Un poquito de historia
- 3) BiSSE (Maddison et al., 2007)
 - Anatomía de un BiSSE
 - Derivemos una likelihood!
 - SSE super-bloom: MuSSE, quaSSE, GeoSSE.
- 4) BiSSE's controversy
 - Replication (Maddison & FitzJohn, 2015)
 - Model inadequacy (Rabosky & Goldberg, 2015)

1) Motivación

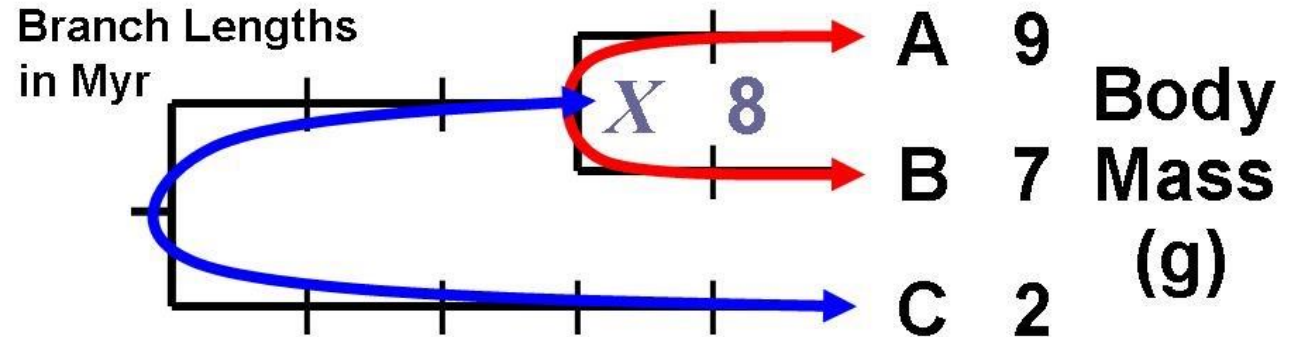
- Comparar diferencias/correlaciones en caracteres (continuos) entre clados.
- ¿Qué efecto tiene un carácter en la diversificación de un clado?
- ¿Cuál es la historia evolutiva de un carácter?



2) Un poquito de historia

PIC (Phylogenetic independent contrasts)

- “Corregir” el efecto de la no-independencia de las muestras.
- Popular para aplicar pruebas estadísticas a datos filogenéticos, comunmente continuos.



Identify and Compute Independent Contrasts
Compute square roots of sums of
(corrected) branch lengths = S.D.

Contrast	Value	S.D.
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A-B	2	2
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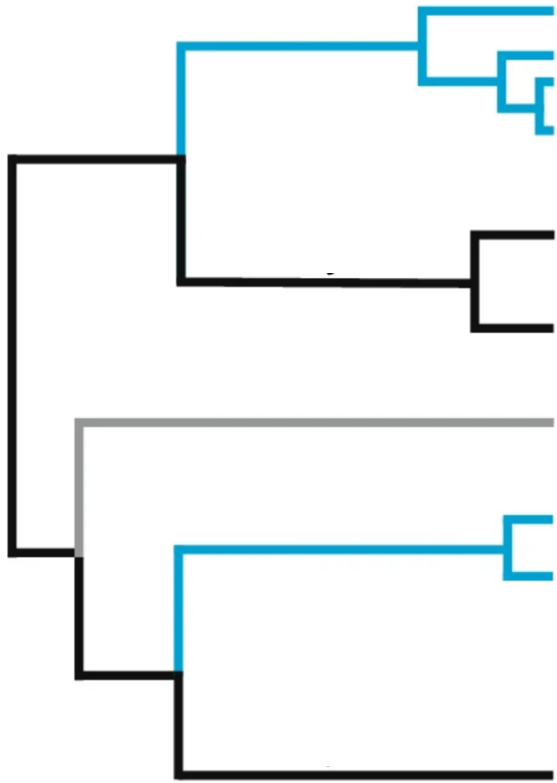
X-C	6	3
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Standardized Contrast

1

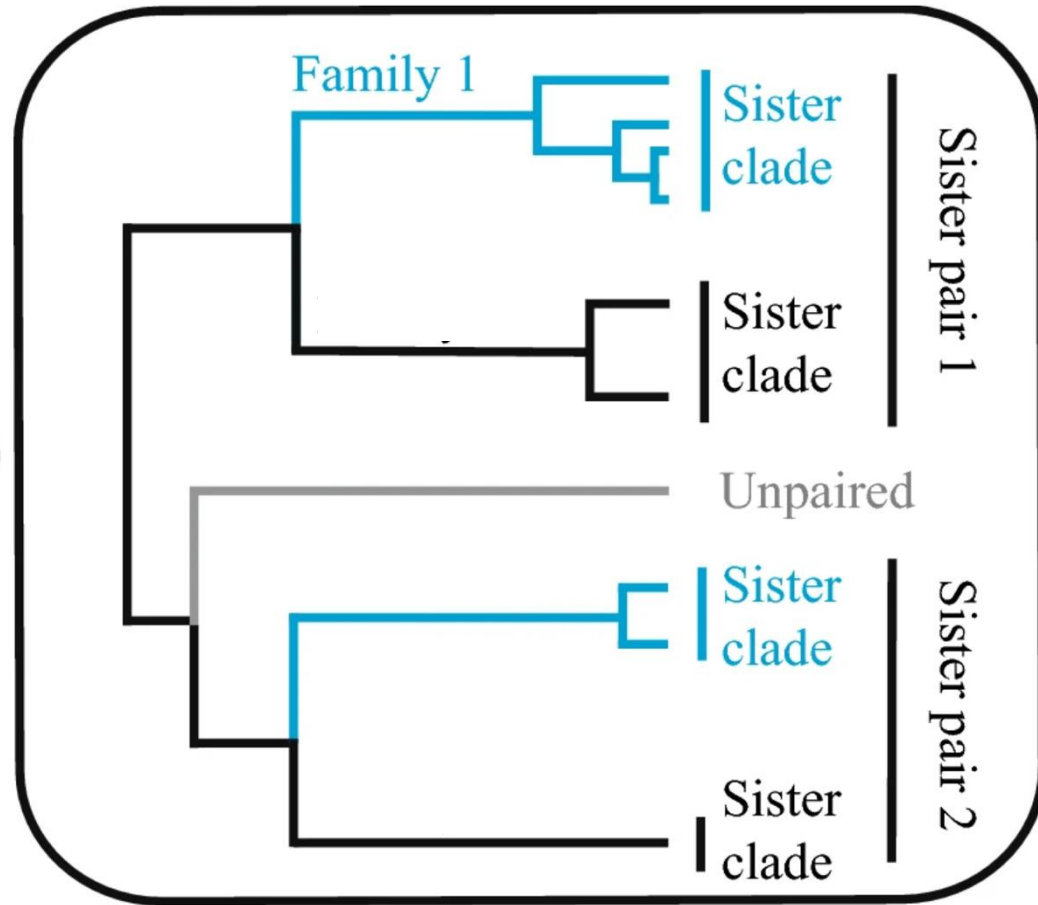
2

Sister-clade comparisons



- Comparar tasas de diversificación entre grupos.

Sister-clade comparisons



- Comparar tasas de diversificación entre grupos.

3) Anatomía de un modelo BiSSE

Binary State-dependent Speciation and Extinction

Recordatorio: Modelo BD (birth-death)

Birth-death model.

λ - especiación

μ - extinción

ρ - sampling fraction

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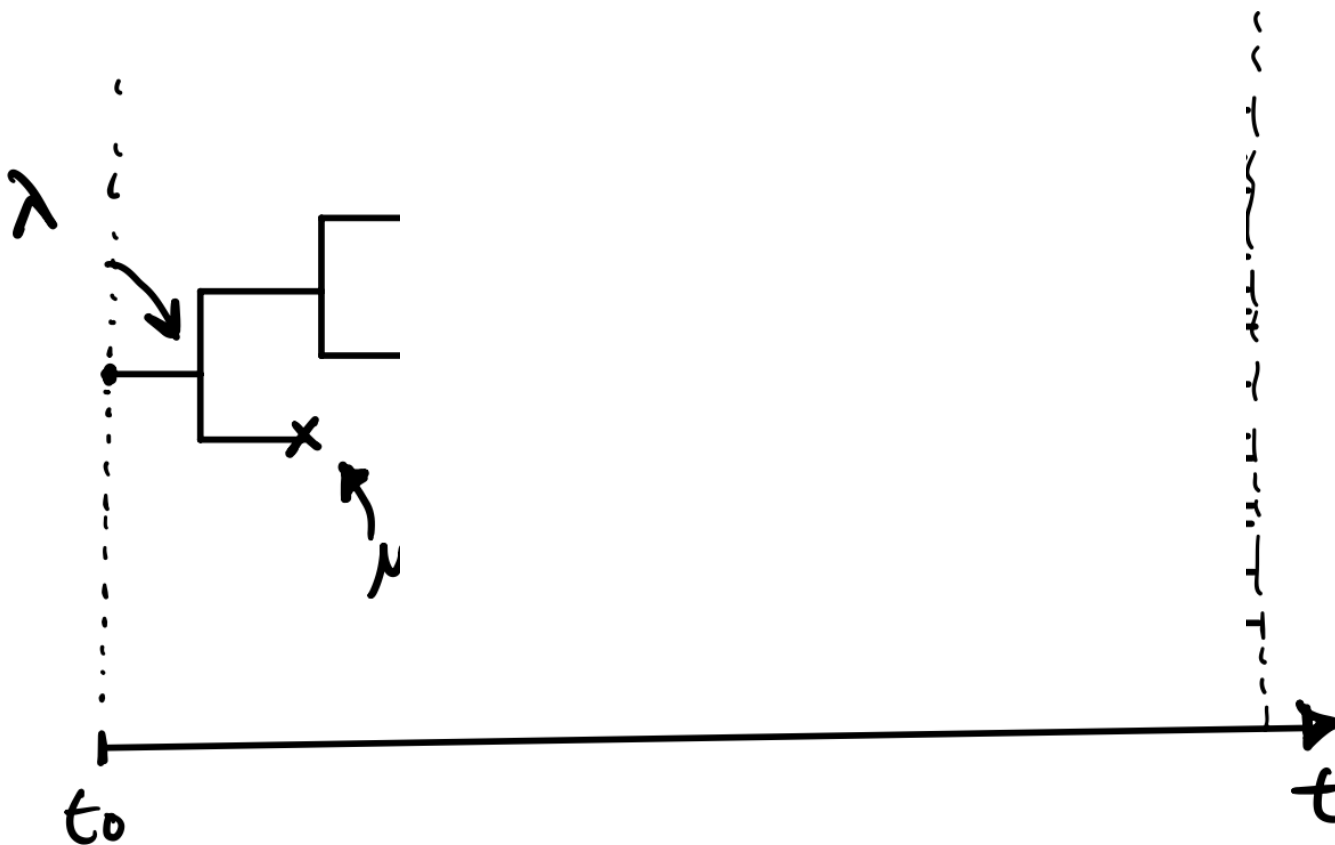
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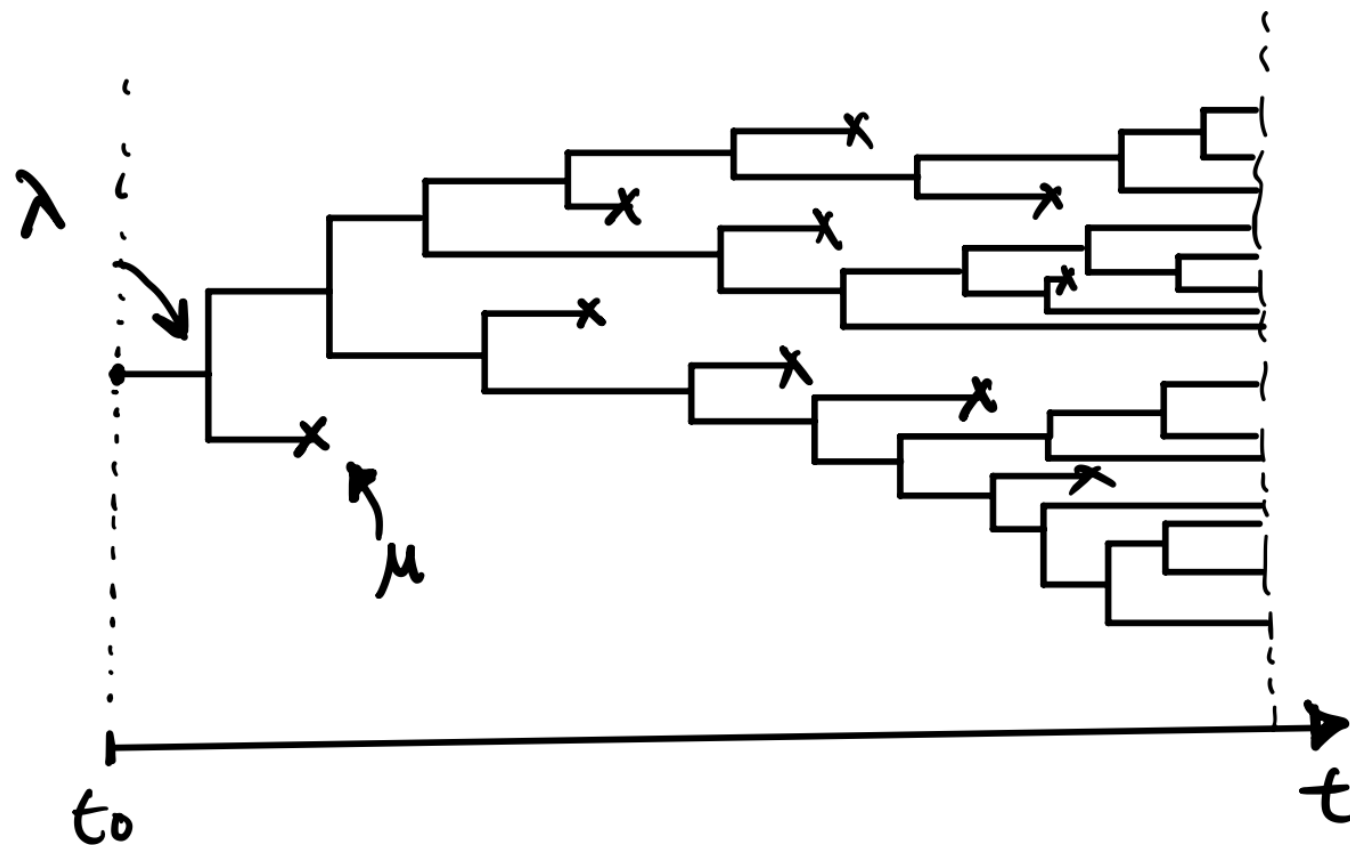
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Birth-death model.

λ - especiación

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Modelo BiSSE

Bi SSE
Binary State Speciation
State and extinction

Estado 0

λ_0

μ_0

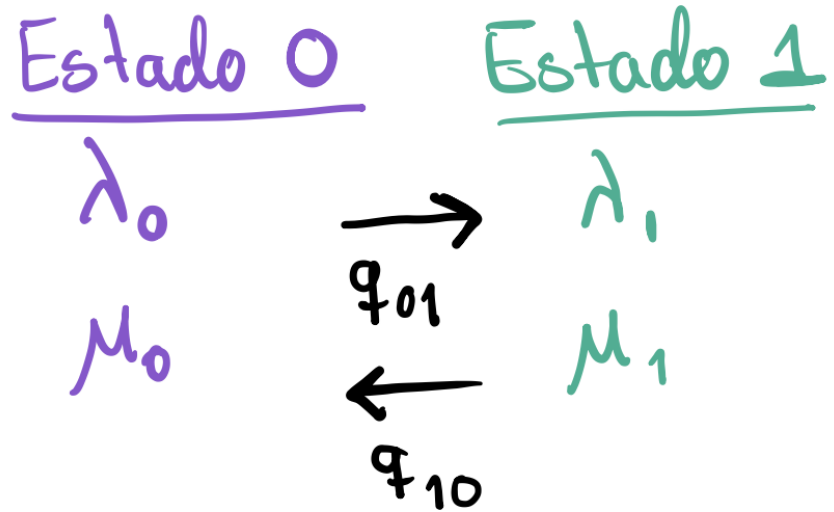
Estado 1

λ_1

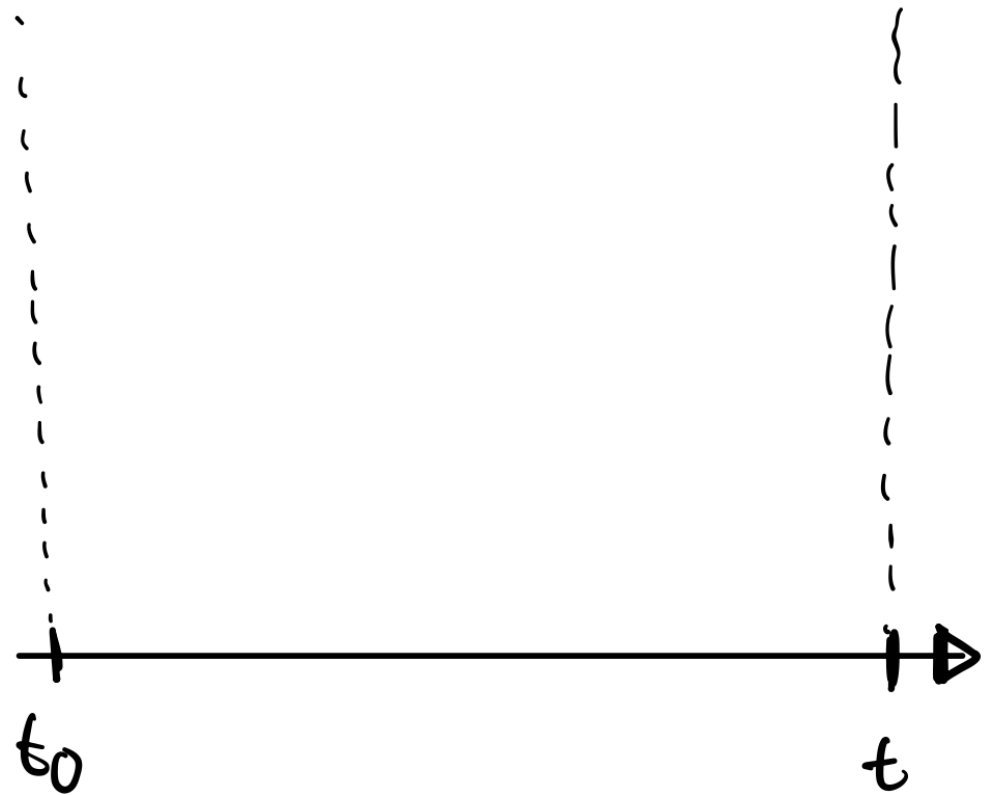
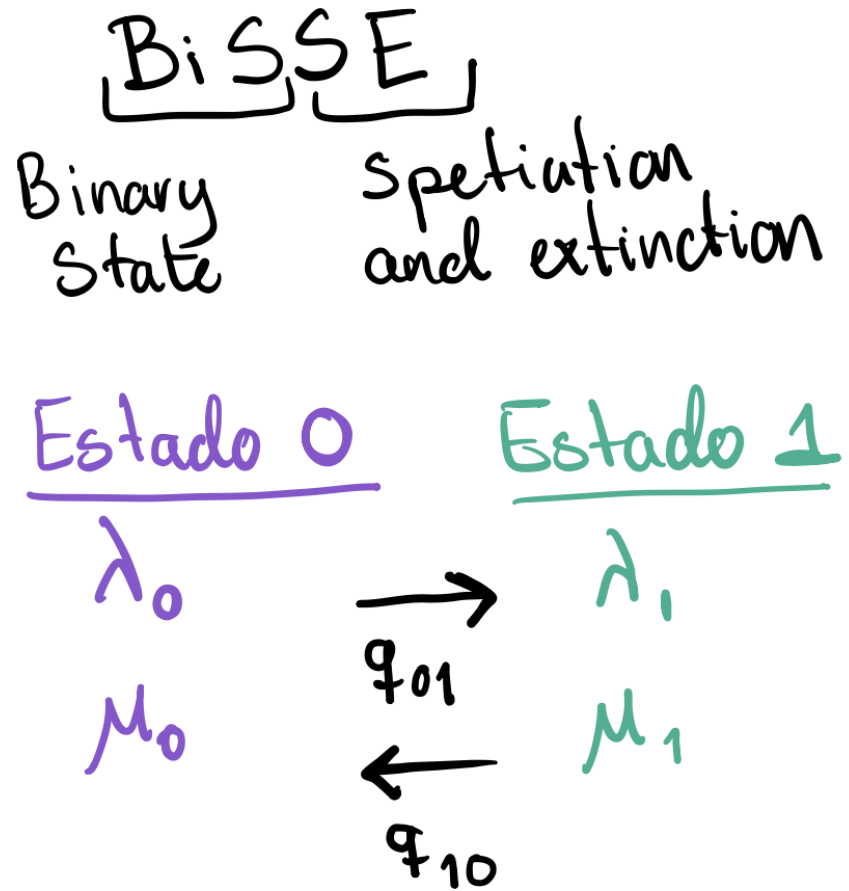
μ_1

Modelo BiSSE

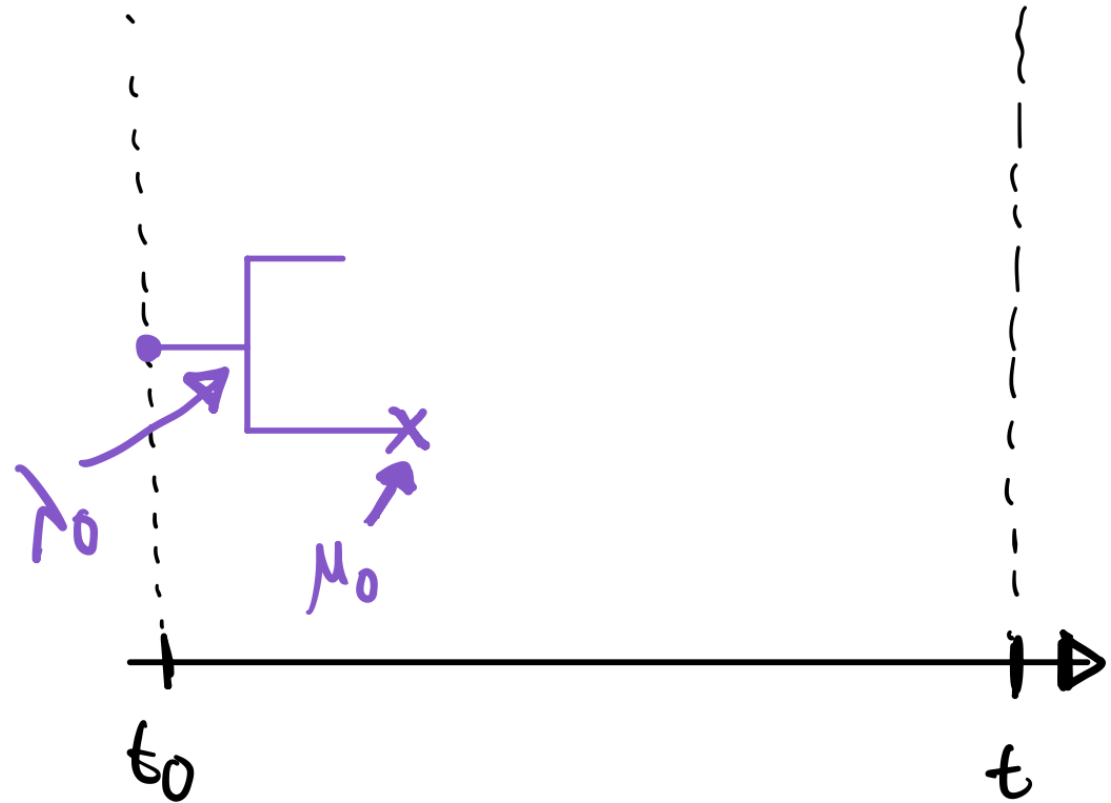
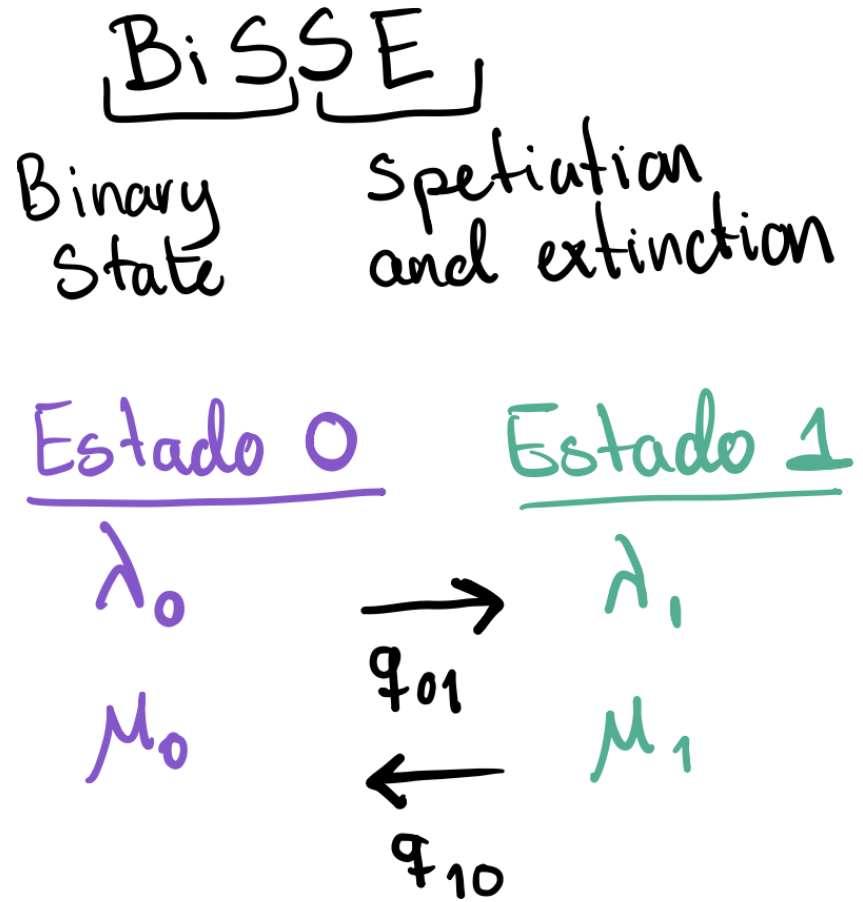
BiSSE
Binary State Speciation
State and extinction



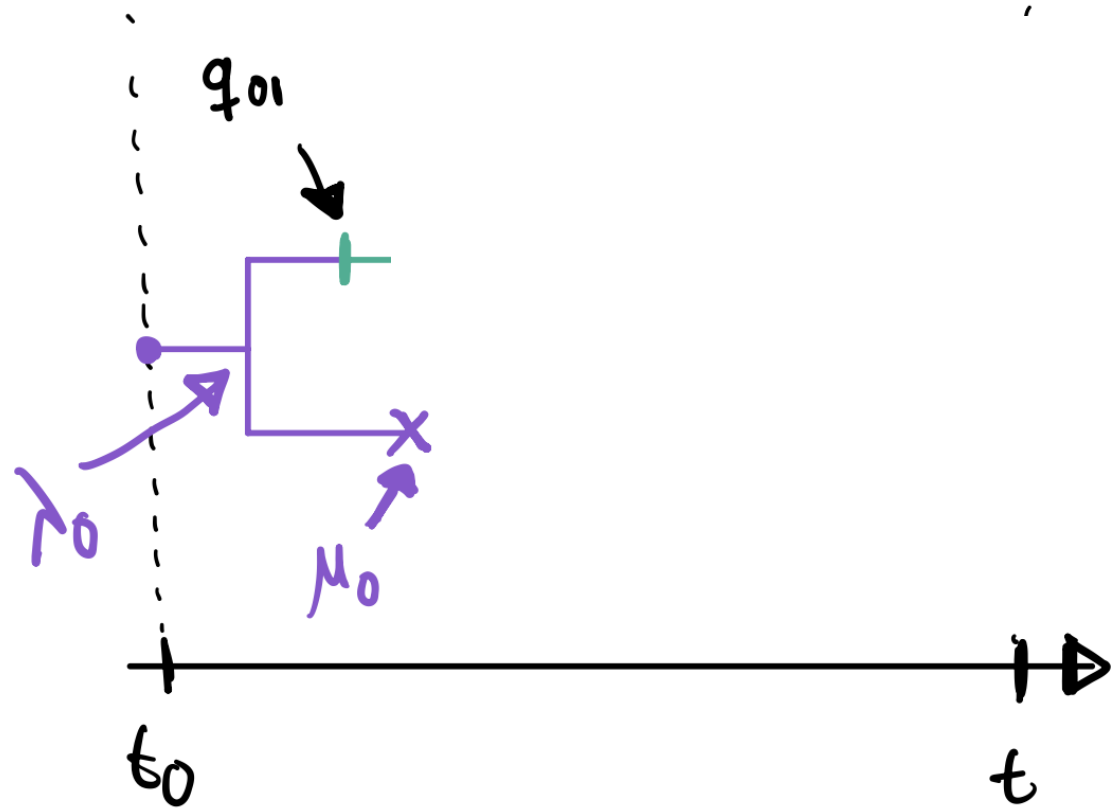
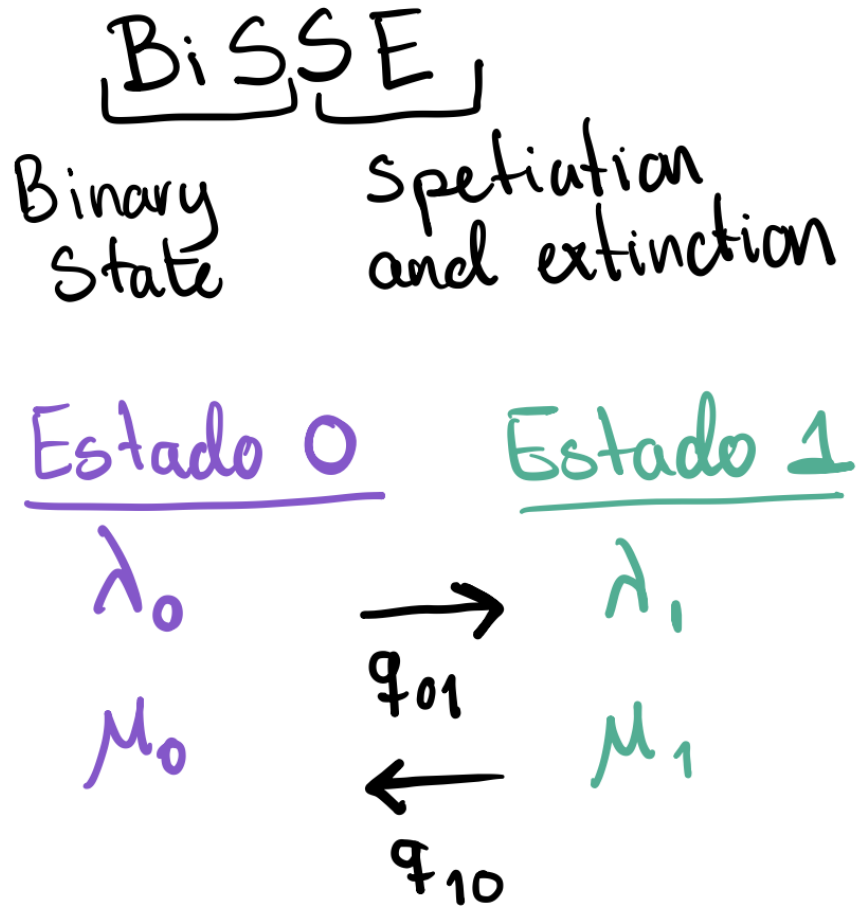
Modelo BiSSE



Modelo BiSSE



Modelo BiSSE



Modelo BiSSE

BiSSE

Binary State Speciation and extinction

Estado 0

 λ_0

μ_0

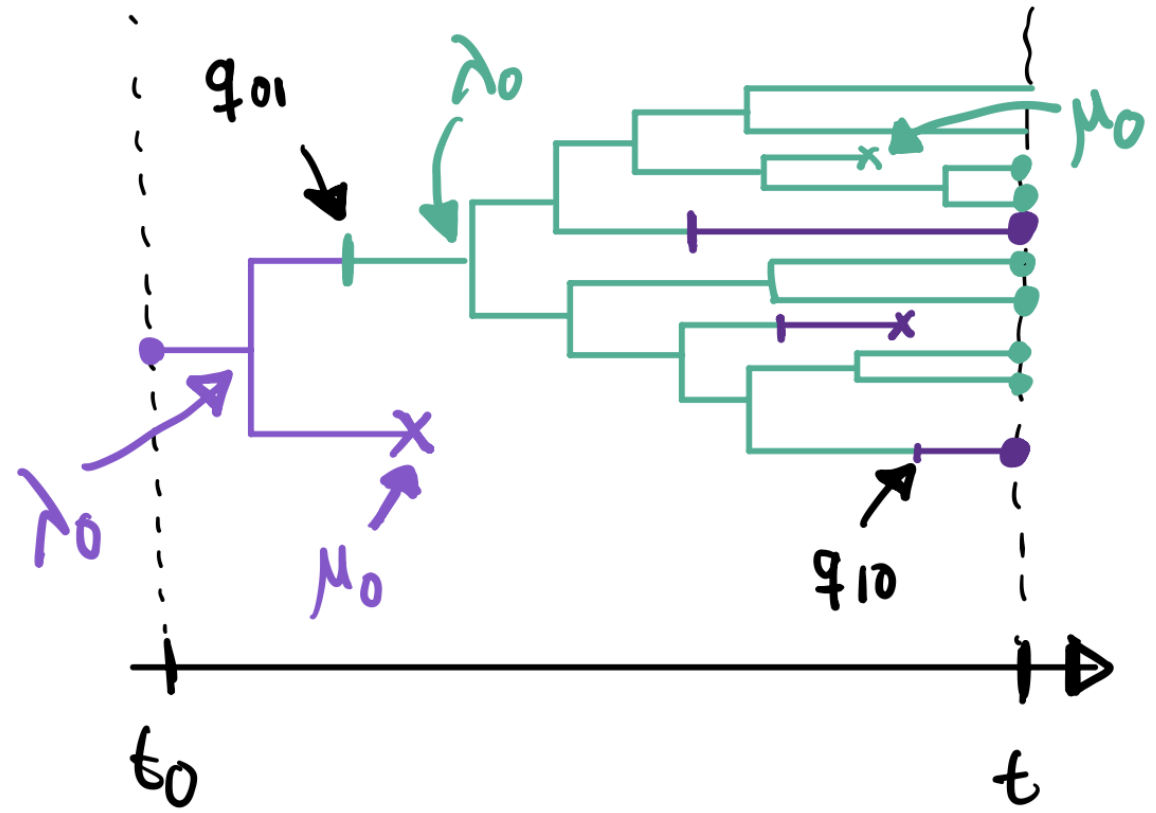
Estado 1

 λ_1 μ_1 

901

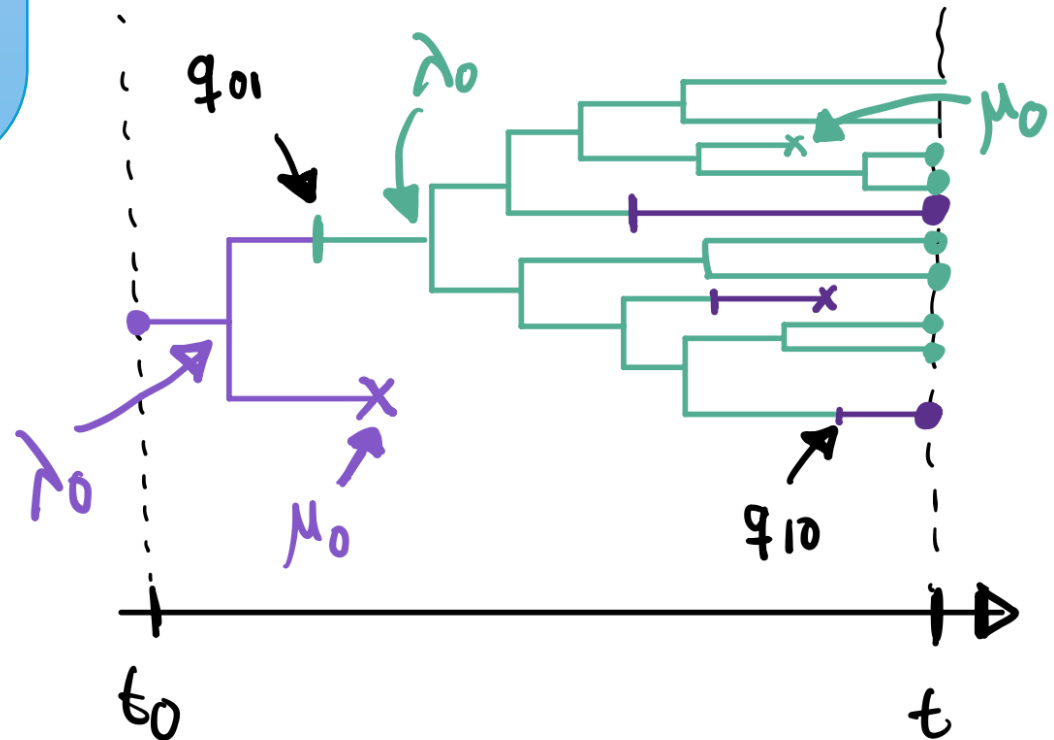
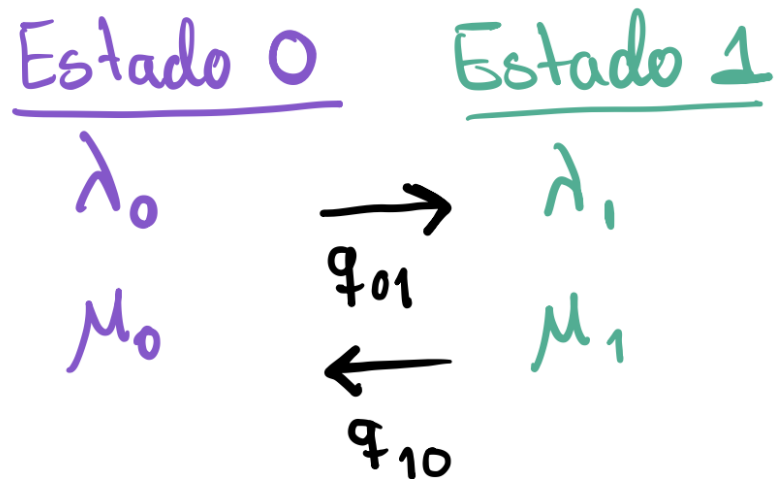


910



Modelo BiSSE

- 1) Si fijo los valores de los parametros y simulo 1000 árboles, ¿van a ser todos iguales?
- 2) Usando simulaciones, ¿cómo obtendría la probabilidad de un árbol?
- 3) Usando simulaciones, ¿cómo infiero los estados ancestrales?
- 4) ¿Estados ancestrales y mapas estocásticos son lo mismo?



¿Cómo hacemos inferencia?

$$p(\theta \mid \text{data}) = \frac{p(\text{data} \mid \theta) \cdot p(\theta)}{p(\text{data})}$$

Diagram illustrating the components of the Bayesian inference formula:

- Posterior**: $p(\theta \mid \text{data})$
- Likelihood**: $p(\text{data} \mid \theta)$
- Prior**: $p(\theta)$
- Normalization**: $p(\text{data})$

¿Cómo hacemos inferencia?

$$p(\theta \mid \text{data}) = \frac{p(\text{data} \mid \theta) \cdot p(\theta)}{p(\text{data})}$$

Diagram illustrating the components of the posterior distribution formula:

- Posterior: $p(\theta \mid \text{data})$
- Likelihood: $p(\text{data} \mid \theta)$
- Prior: $p(\theta)$
- Normalization: $p(\text{data})$

$$\frac{dD_{N,i}(t)}{dt} = -(\lambda_i + \mu_i + q_{ij}) D_{N,i}(t) + q_{ij} D_{N,j}(t) + 2\lambda_i E_i(t) D_{N,i}(t)$$

$$\frac{dE_i(t)}{dt} = \mu_i - (\lambda_i + \mu_i + q_{ij}) E_i(t) + q_{ij} E_j(t) + \lambda_i E_i(t)^2$$

Atributos de un modelo SSE

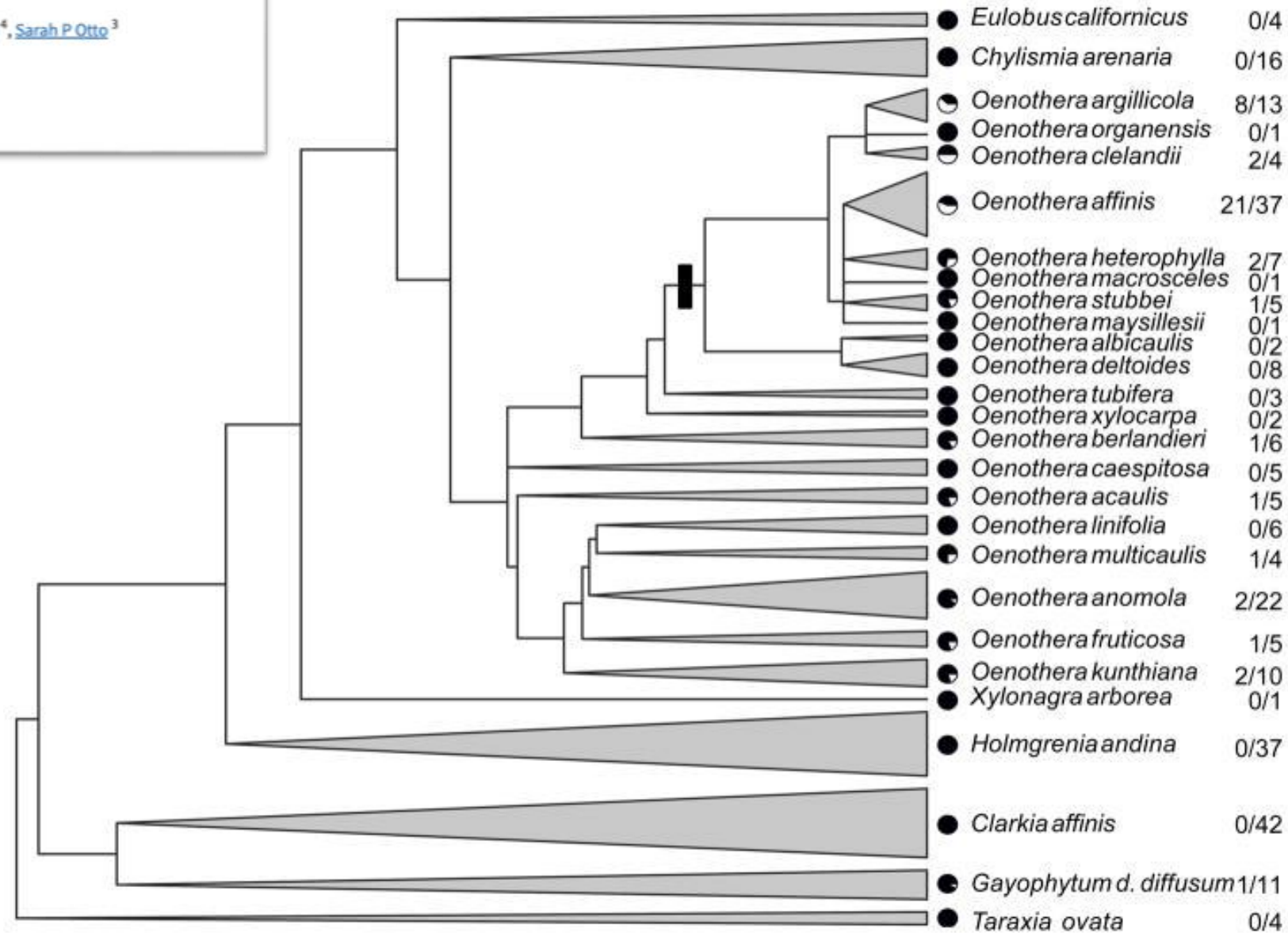
- Modela simultaneamente (1) el proceso de diversificación y (2) la evolución de un carácter (por medio de un CTMC).
- ¡Toma en cuenta la posibilidad de eventos no observados a la hora de calcular la verosimilitud!
- Permite separar el efecto de la especiación del efecto de la extinción (en contraste con sister-clade comparisons).

LOSS OF SEXUAL RECOMBINATION AND SEGREGATION IS ASSOCIATED WITH INCREASED DIVERSIFICATION IN EVENING PRIMROSES

Marc T. J. Johnson^{1,2,*}, Richard G. FitzJohn³, Stacey D. Smith^{4,5}, Mark D. Rausher⁴, Sarah P. Otto³

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PMCID: PMC3202692 NIHMSID: NIHMS309207 PMID: [22023588](https://pubmed.ncbi.nlm.nih.gov/22023588/)

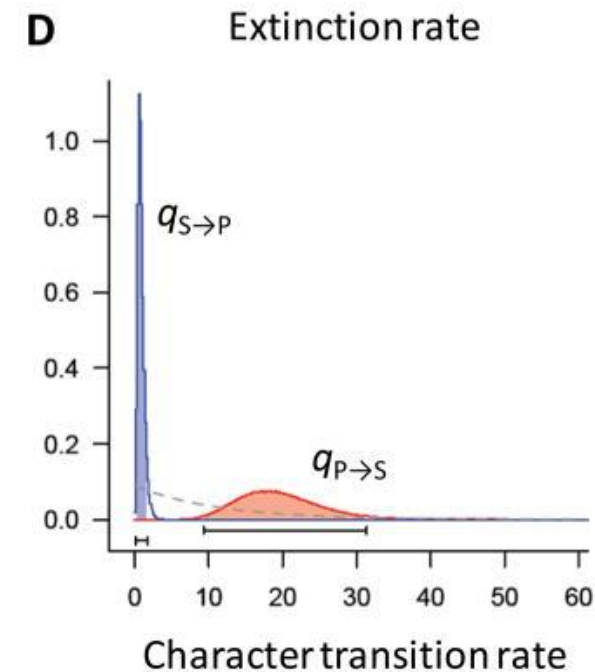
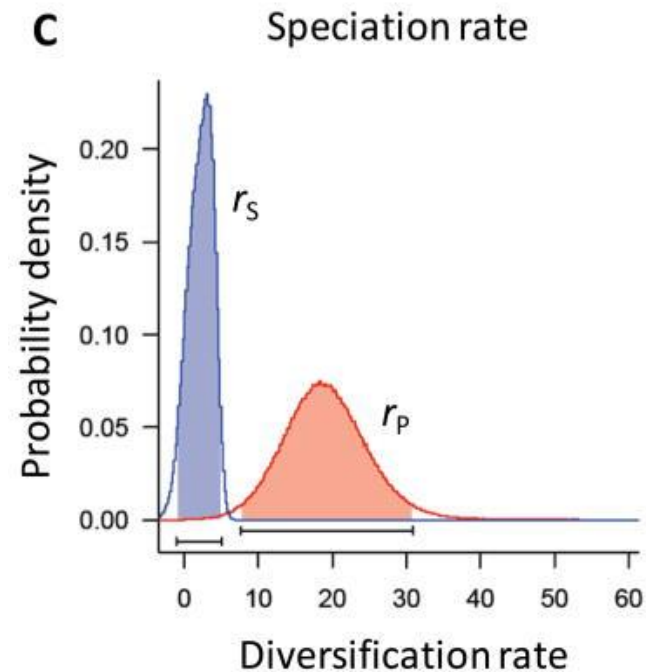
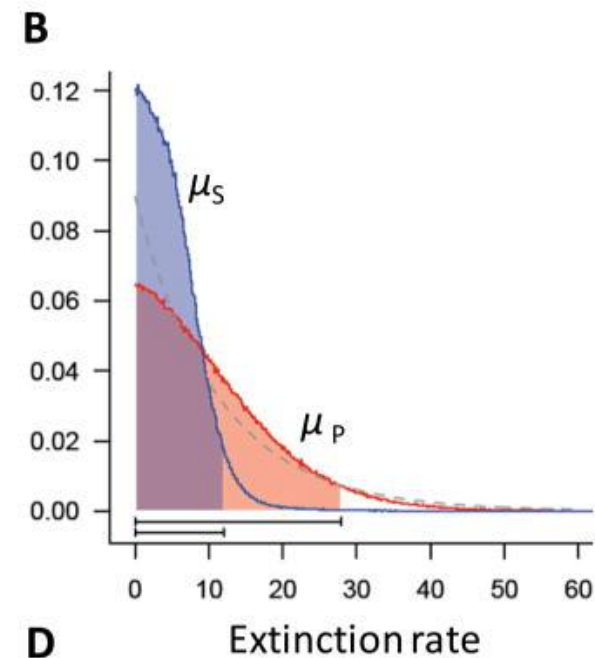
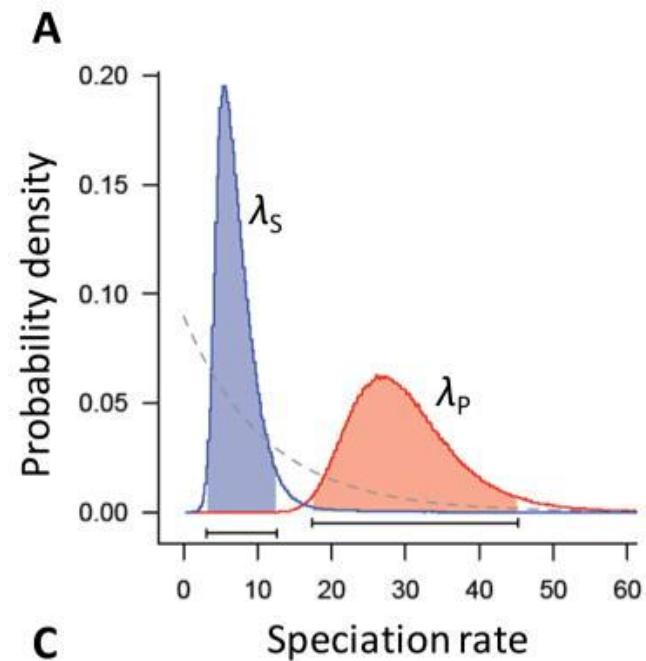


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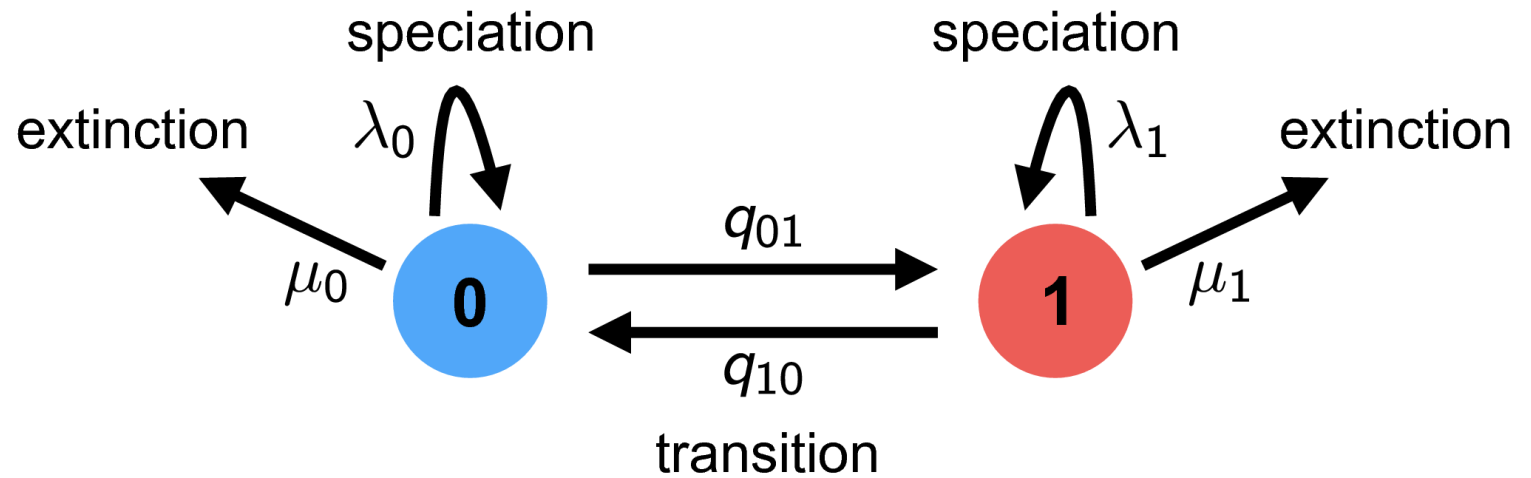
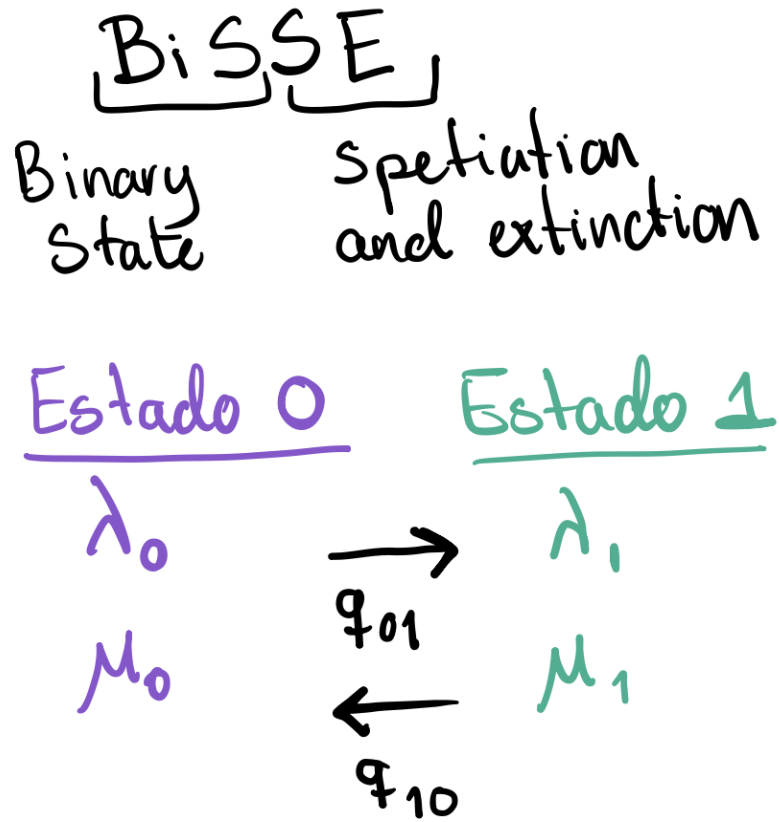
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Modelo BiSSE



BiSSE

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Estimating a Binary Character's Effect on Speciation and Extinction

WAYNE P. MADDISON,^{1,2,3,4} PETER E. MIDFORD,¹ AND SARAH P. OTTO^{1,2}

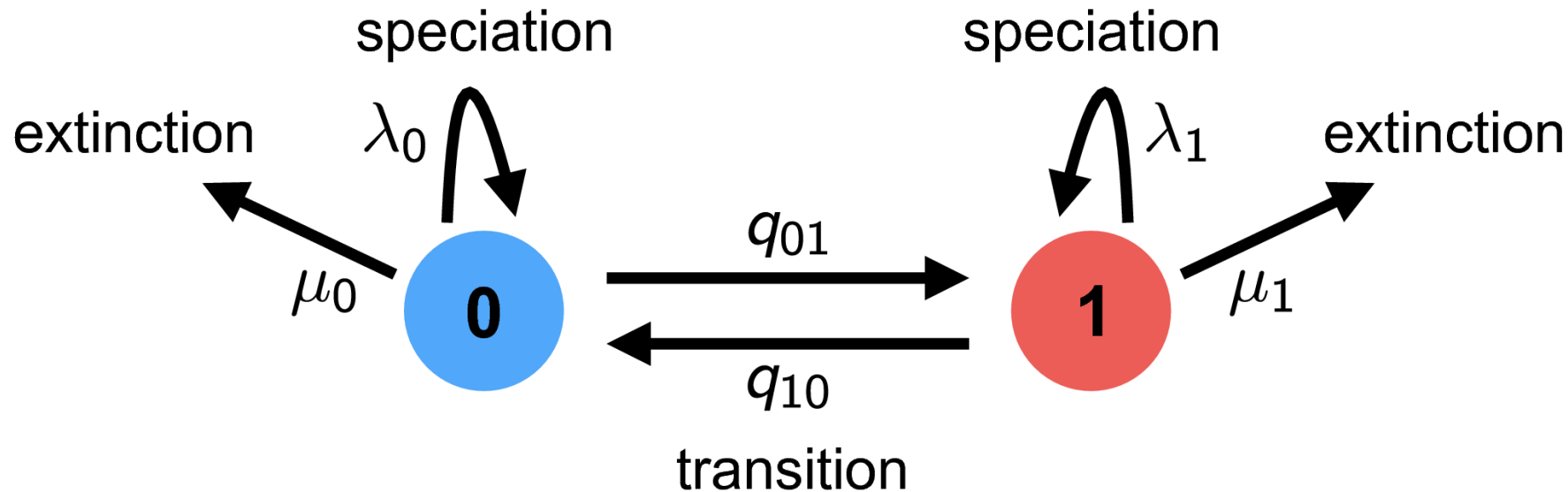
¹Department of Zoology, University of British Columbia, Vancouver, BC V6T 1Z4, Canada; E-mail: wmaddisn@interchange.ubc.ca (W.P.M.)

²Biodiversity Research Centre, University of British Columbia, Vancouver, BC V6T 1Z4, Canada

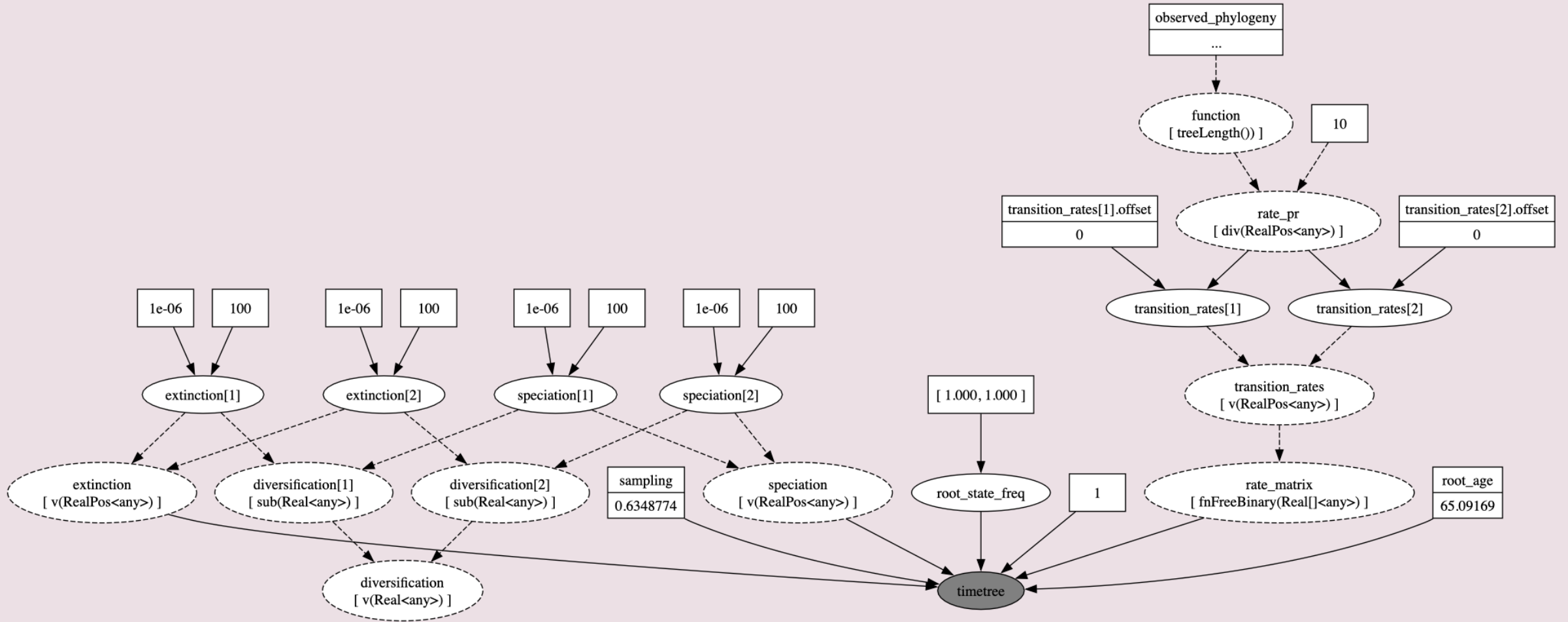
³Department of Botany, University of British Columbia, Vancouver, BC V6T 1Z4, Canada

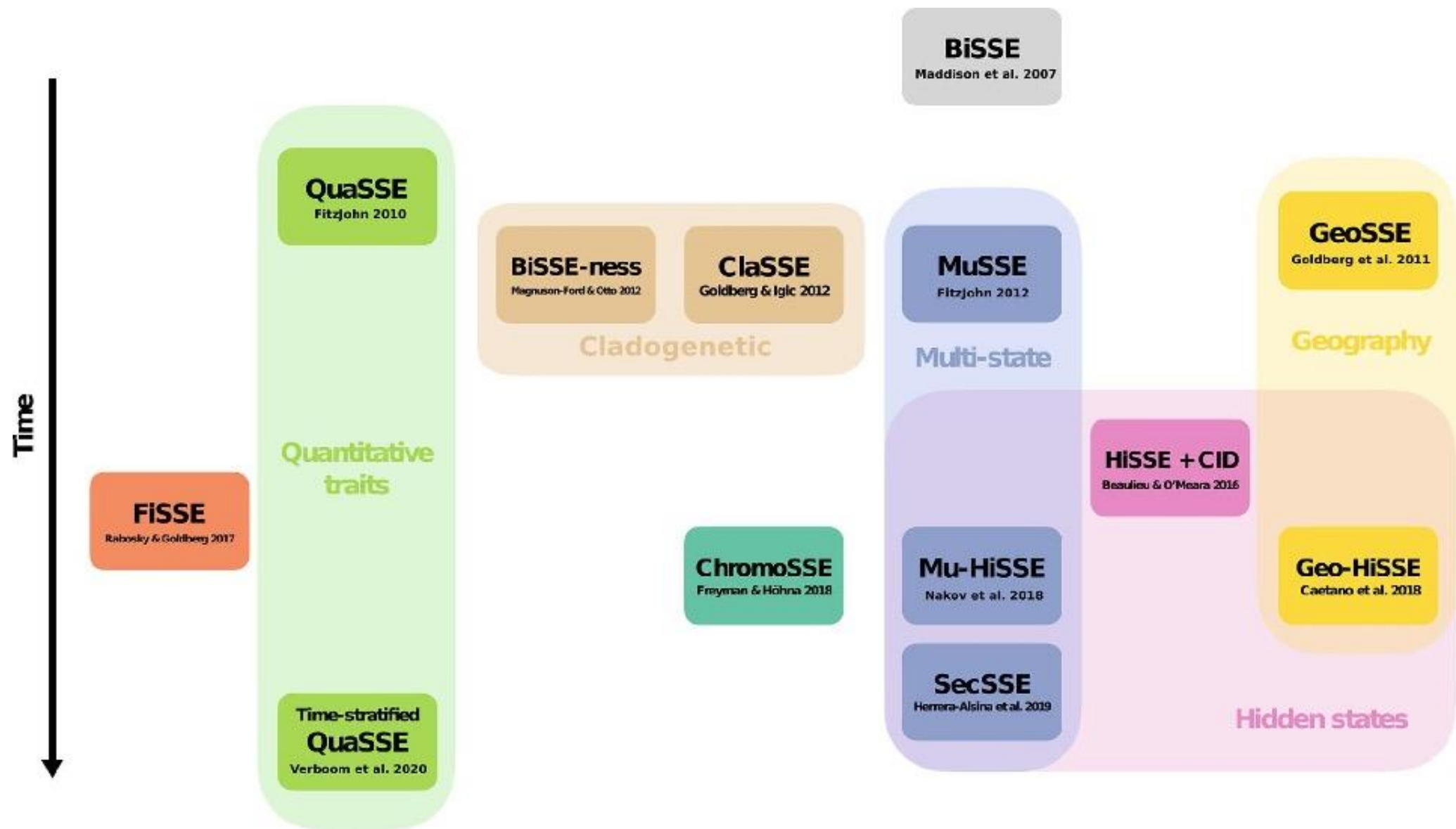
⁴Wissenschaftskolleg zu Berlin, Wallotstraße 19, Berlin 14193, Germany

Abstract.—Determining whether speciation and extinction rates depend on the state of a particular character has been of long-standing interest to evolutionary biologists. To assess the effect of a character on diversification rates using likelihood methods requires that we be able to calculate the probability that a group of extant species would have evolved as observed, given a particular model of the character's effect. Here we describe how to calculate this probability for a phylogenetic tree and a two-state (binary) character under a simple model of evolution (the "BiSSE" model, binary-state speciation and extinction). The model involves six parameters, specifying two speciation rates (rate when the lineage is in state 0; rate when in state 1), two extinction rates (when in state 0; when in state 1), and two rates of character state change (from 0 to 1, and from 1 to 0). Using these probability calculations, we can do maximum likelihood inference to estimate the model's parameters and perform hypothesis tests (e.g., is the rate of speciation elevated for one character state over the other?). We demonstrate the application of the method using simulated data with known parameter values. [Birth-death process; branching process; cladogenesis; extinction; key innovation; macroevolution; phylogeny; speciation; speciose; statistical inference.]



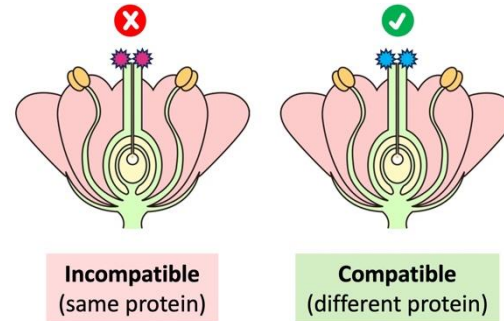
BiSSE en revbayes! (El viernes)





ClaSSE (Cladogenetic events)

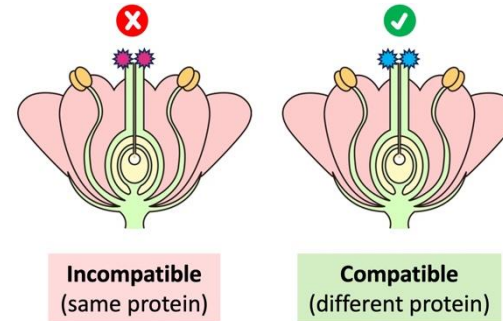
Evolución de la
compatibilidad sexual
en Solanaceae.



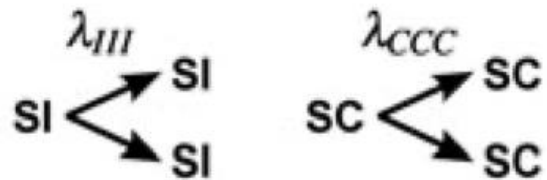
ClaSSE

(Cladogenetic events)

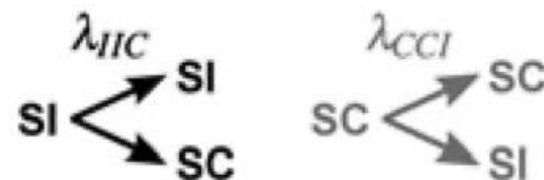
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Cladogenesis, no state change
(BiSSE & ClaSSE)



Cladogenetic state change
(ClaSSE only)



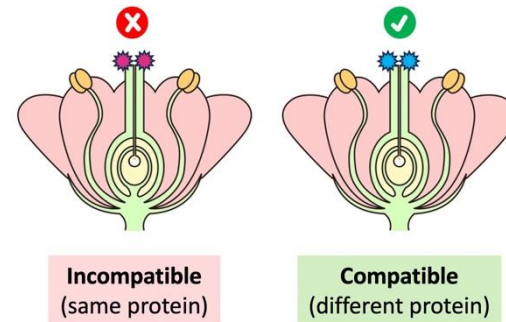
Anagenetic state change (BiSSE & ClaSSE)



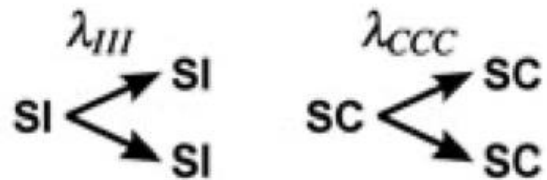
ClaSSE

(Cladogenetic events)

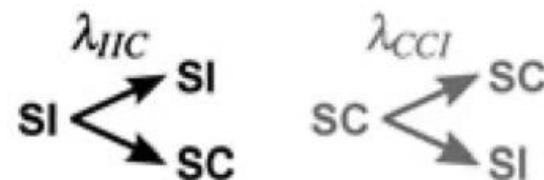
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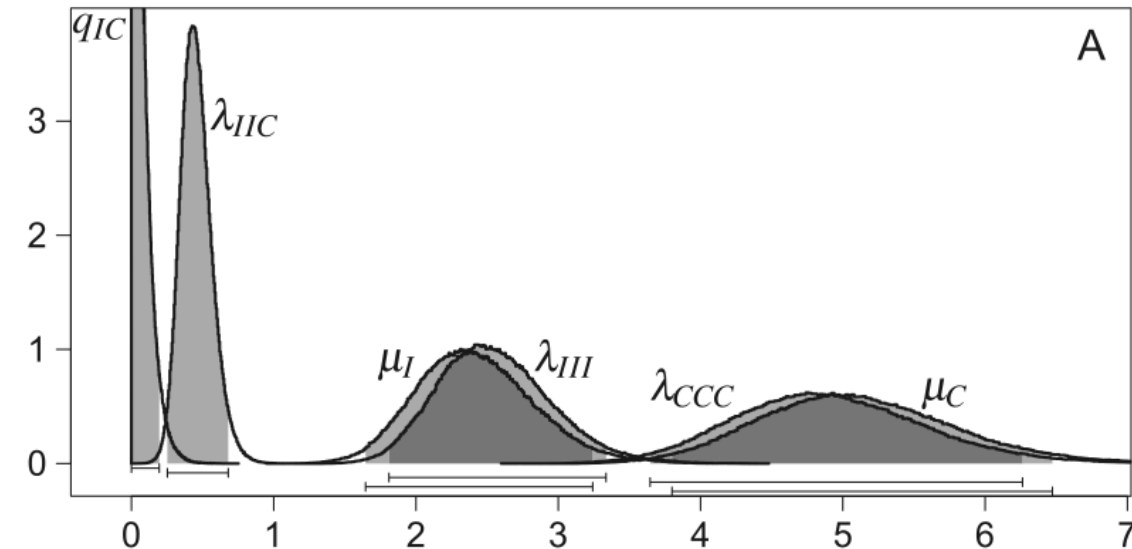
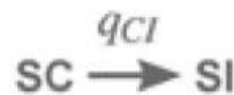
Cladogenesis, no state change
(BiSSE & ClaSSE)



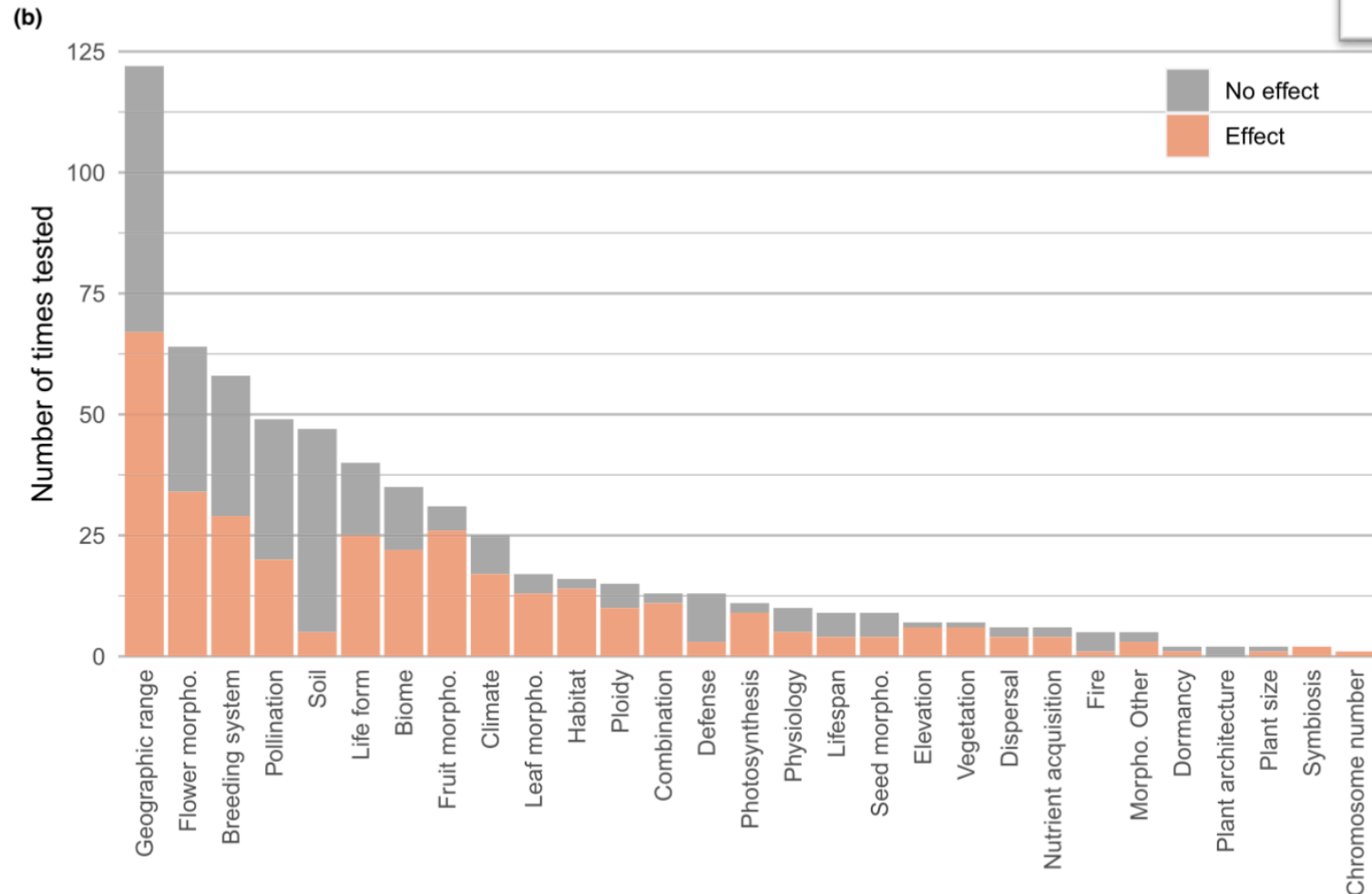
Cladogenetic state change
(ClaSSE only)



Anagenetic state change (BiSSE & ClaSSE)



La popularidad de los modelos SSE



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DOI: 10.1111/ele.14179

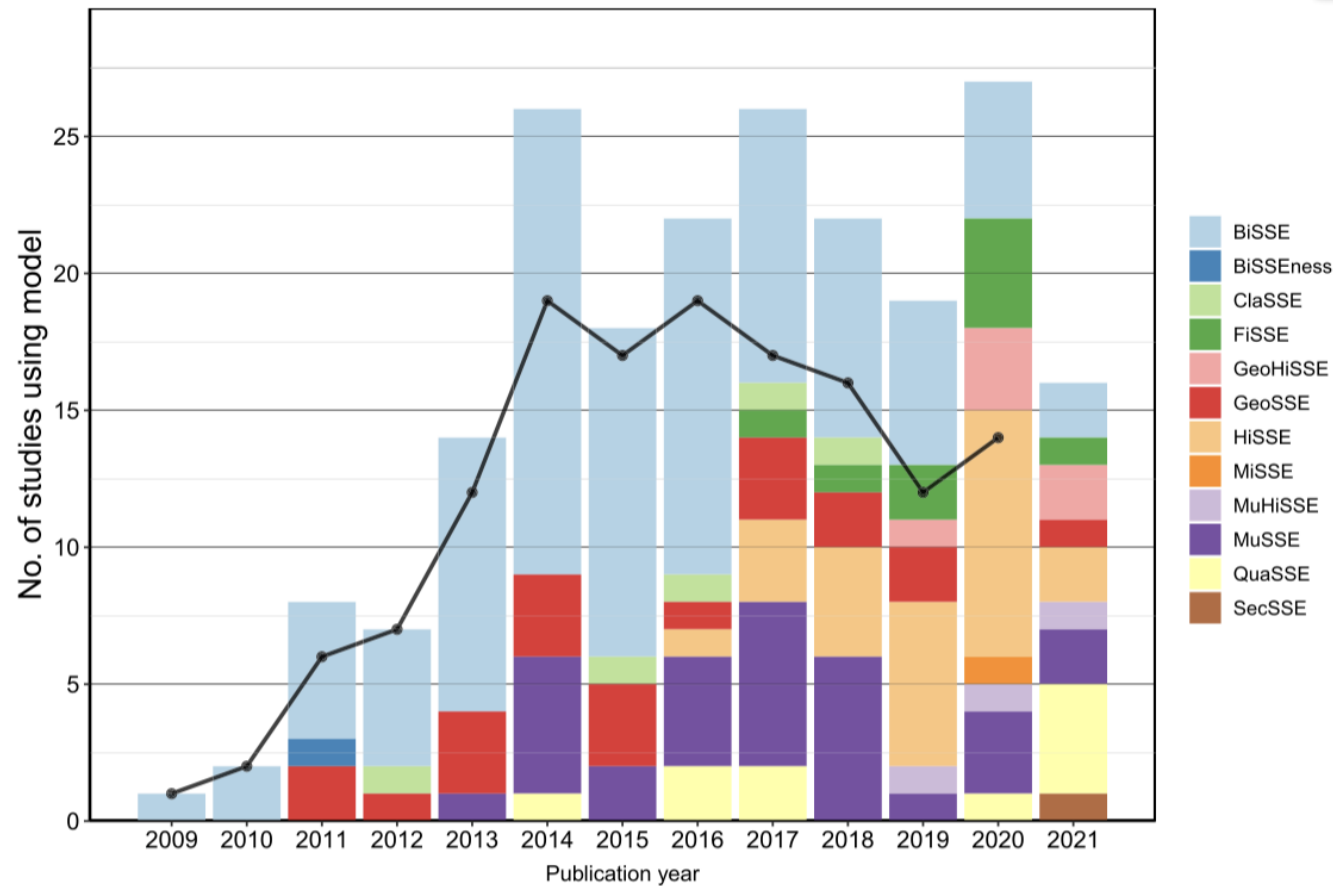
SYNTHESIS

ECOLOGY LETTERS WILEY

Trait-dependent diversification in angiosperms: Patterns, models and data

Andrew J. Helmstetter¹ | Rosana Zenil-Ferguson² | Hervé Sauquet^{3,4} | Sarah P. Otto⁵ | Marcos Méndez⁶ | Mario Vallejo-Marin⁷ | Jürg Schönenberger⁸ | Concetta Burgarella⁹ | Bruce Anderson¹⁰ | Hugo de Boer¹¹ | Sylvain Glémin^{7,12} | Jos Käfer^{13,14}

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La popularidad de los modelos SSE

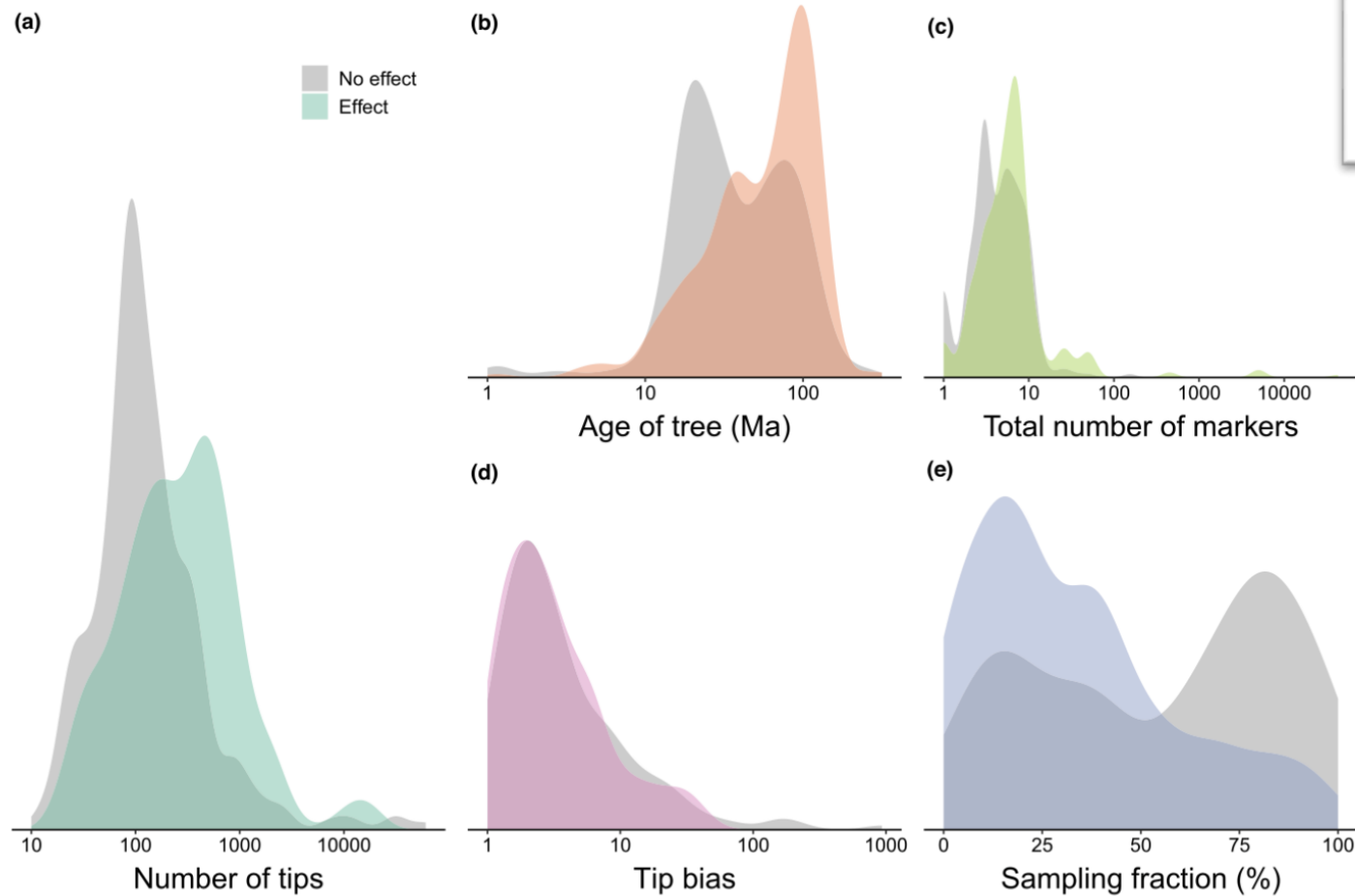


FIGURE 4 A set of densities depicting the distribution of values for five data set properties in SSE models that infer trait dependent diversification (coloured densities) and those that do not (grey densities). The data set properties shown are (a) number of tips in the phylogenetic tree used with the SSE model (data taken from $n = 621$ models), (b) the age of the tree used with the SSE model ($n = 529$), (c) the total number of genetic markers (nuclear + plastid + mitochondrial) used to infer the phylogenetic tree used with the SSE model ($n = 615$), (d) the tip bias, here calculated as the largest tip frequency divided by the smallest ($n = 426$) and (e) the global sampling fraction ($n = 616$).

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